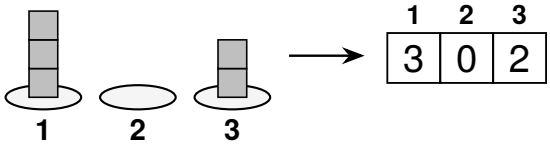
Both players have the same number of cubes. Behind your folder, put all your cubes on the three hills of your mat. This is your plan. A hill can stay empty. Lift the folders at the same time. On each hill, the player with more cubes wins the hill and puts a token on it. If the cubes are equal, nobody wins that hill. The player who wins more hills wins the game. If both players win the same number of hills, the game is a tie. A plan is written as three numbers in a strip: the cubes on hills 1, 2 and 3.



Problem 1: Play the game with 6 cubes each. Is there a plan that nobody can beat? To test a plan, one player shows it and the others try to beat it.

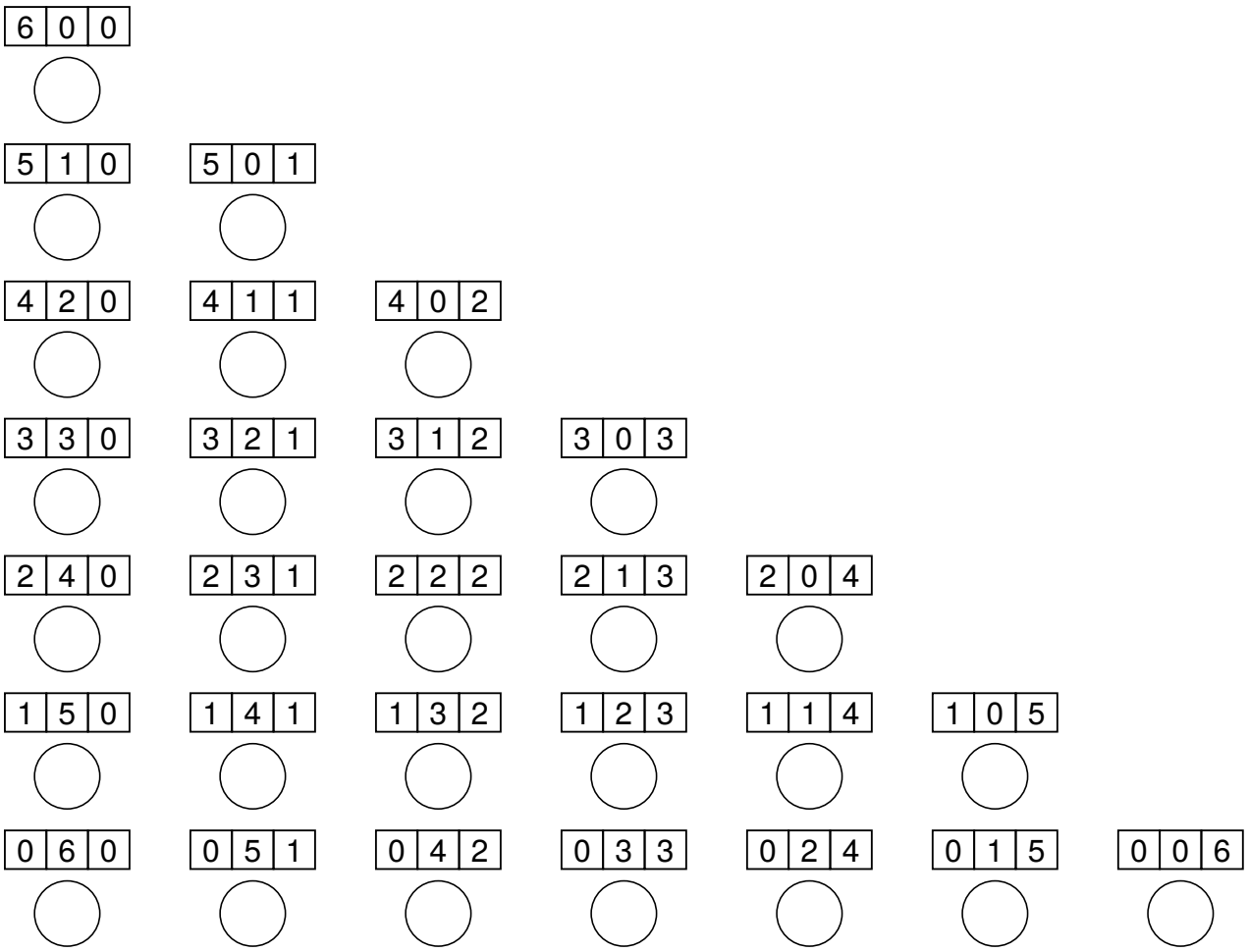
plan shown | plan that beats it | plan shown | plan that beats it

Problem 2: Beat each plan below. Then explain a way to beat any plan with 6 or more cubes that your partner names.

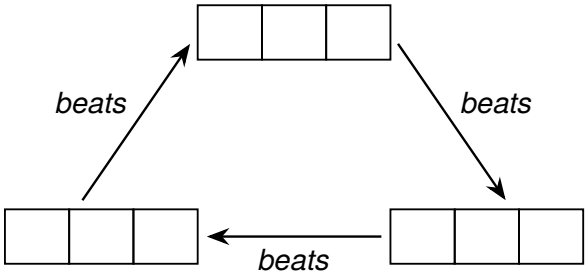
6 cubes	2 2 2		1 4 1	
7 cubes	3 2 2		1 1 5	
9 cubes	3 3 3			
10 cubes	1 8 1			
20 cubes	7 7 6			
100 cubes	34 33 33		50 0 50	

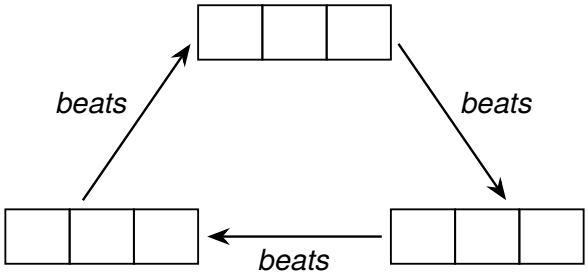
Problem 3: For which numbers of cubes is there a plan that nobody can beat? Explain.

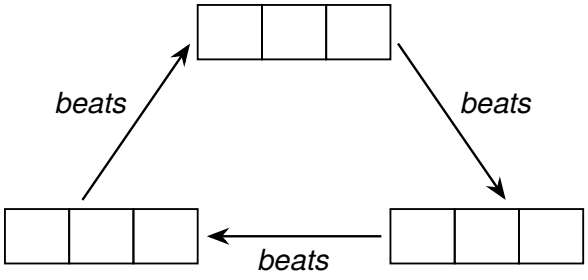
Problem 4: There are 28 plans with 6 cubes. For each one, find how many of the 28 plans it beats. Share the work. Which plans beat the most? Which beat the fewest?

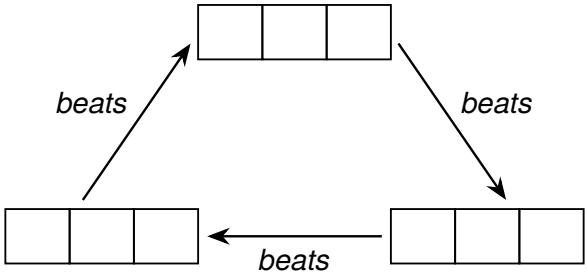


Problem 5: Find three plans with 6 cubes so that each plan beats the next one around the circle. Which plans with 6 cubes can be part of a circle like this, and which cannot? Explain why.

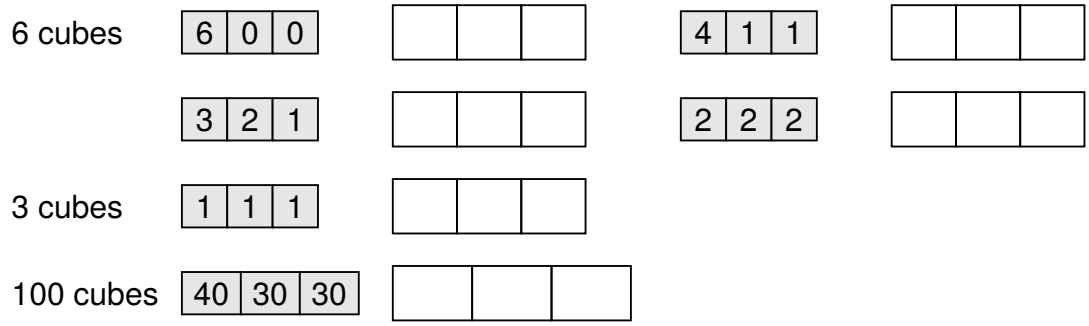








Problem 6: Now the two players may have different numbers of cubes. What is the fewest cubes that can beat each plan below? Find a rule that works for any plan, and explain why it works.



Problem 7: Your partner has 6 cubes. What is the fewest cubes you need for a plan that your partner cannot beat? For a plan that beats every plan your partner could make?

Problem 8: Ben and Mia each choose a plan with 6 cubes by rolling a die and using the list below. You know their lists but not their rolls. For each of them, look for a plan with 6 cubes that beats more of their six plans than it loses to. If there is none, explain why.

	Ben	Mia
	3 2 1	4 2 0
	3 1 2	4 0 2
	2 3 1	2 4 0
	2 1 3	2 0 4
	1 3 2	0 4 2
	1 2 3	0 2 4